

SQL COMPLETE GUIDE — PART 1

FOUNDATION → DQL/DML/DDD/DCL/TCL → FUNCTIONS → AGGREGATION → JOINS → SUBQUERIES → CTEs

1. SQL Fundamentals

SQL — Definition: Structured Query Language used to create, retrieve, modify, control and analyze data in relational databases.

RDBMS — Definition: Software that stores related data in tables and provides SQL/database management capabilities.

Database — Definition: An organized collection of data and database objects.

Schema — Definition: A logical namespace/container for database objects; in MySQL, database and schema are commonly interchangeable.

Table — Definition: A database object containing rows and columns.

Row / Record — Definition: One complete record in a table.

Column / Field — Definition: One attribute of a table.

Data Type — Definition: Defines what values a column can store, such as INT, VARCHAR, DATE or DECIMAL.

Alias — Definition: A temporary name for a table or column in a query.

Scenario: E-commerce has Customers, Products, Orders, Order_Items and Payments. SQL retrieves, joins, filters and summarizes this data.

2. SQL Command Categories

Category	Full form	Commands	Purpose
DQL	Data Query Language	SELECT	Read data
DML	Data Manipulation Language	INSERT, UPDATE, DELETE, MERGE	Change data
DDL	Data Definition Language	CREATE, ALTER, DROP, TRUNCATE, RENAME	Change structure
DCL	Data Control Language	GRANT, REVOKE	Permissions
TCL	Transaction Control Language	COMMIT, ROLLBACK, SAVEPOINT	Transactions

Memory: DQL read → DML rows → DDL structure → DCL access → TCL transactions.

3. DDL

DDL — Definition: Commands that create, modify or remove database objects.

Command	Definition	Example
CREATE	Creates an object	CREATE TABLE
ALTER	Changes structure	ADD/DROP/MODIFY column
DROP	Removes an object	DROP TABLE
TRUNCATE	Removes all rows but keeps structure	TRUNCATE TABLE
RENAME	Changes object name where supported	RENAME TABLE

```
CREATE DATABASE CompanyDB;
CREATE SCHEMA Sales;
CREATE TABLE Employees(
  Employee_ID INT PRIMARY KEY,
  Name VARCHAR(100),
  Salary DECIMAL(10,2)
);
ALTER TABLE Employees ADD Email VARCHAR(150);
```

4. DML

DML — Definition: Commands used to insert, update and delete rows.

```
INSERT INTO Employees(Employee_ID, Name, Salary)
VALUES(101, 'Rahul', 60000);
```

```
UPDATE Employees SET Salary=65000
WHERE Employee_ID=101;
```

```
DELETE FROM Employees
WHERE Employee_ID=101;
```

MERGE — Definition: A conditional source-target synchronization statement that can insert/update and, in some DBMSs, delete matching data.

5. DQL and Filtering

SELECT — Definition: Retrieves data from tables, views or query results.

WHERE — Definition: Filters individual rows.

DISTINCT — Definition: Removes duplicate result rows.

ORDER BY — Definition: Sorts the result.

LIMIT/TOP/FETCH — Definition: Restricts returned rows; syntax differs by DBMS.

LIKE — Definition: Matches text patterns using wildcards such as % and _.

IN — Definition: Checks whether a value matches a list or subquery.

BETWEEN — Definition: Checks whether a value is within a range.

NULL — Definition: Represents an unknown/missing value; use IS NULL or IS NOT NULL.

```
SELECT DISTINCT Department
FROM Employees
WHERE Salary BETWEEN 50000 AND 80000
ORDER BY Department;
```

6. Operators

Type	Examples	Purpose
Comparison	=, <>, >, <, >=, <=	Compare
Logical	AND, OR, NOT	Combine conditions
Arithmetic	+ - * / %	Calculate
Membership	IN, NOT IN	List/subquery
Range	BETWEEN	Range
Pattern	LIKE	Text
Null	IS NULL	Missing value
Existence	EXISTS	Subquery existence
Quantified	ANY, SOME, ALL	Subquery comparison

7. Logical Query Processing

Logical processing order — Definition: Conceptual order used to understand SQL: FROM/JOIN → WHERE → GROUP BY → HAVING → SELECT → DISTINCT → ORDER BY → LIMIT/FETCH.

8. Aggregate Functions

Aggregate Function — Definition: A function that summarizes multiple rows.

Function	Definition
COUNT()	Counts rows/values

Function	Definition
SUM()	Adds values
AVG()	Calculates average
MIN()	Minimum
MAX()	Maximum

```
SELECT COUNT(*),SUM(Salary),AVG(Salary),
        MIN(Salary),MAX(Salary)
FROM Employees;
```

9. GROUP BY and HAVING

GROUP BY — Definition: Groups rows so aggregate functions can produce one result per group.

HAVING — Definition: Filters groups after aggregation.

```
SELECT Department,COUNT(*) AS cnt,AVG(Salary) AS avg_salary
FROM Employees
GROUP BY Department
HAVING AVG(Salary)>60000;
```

WHERE vs HAVING: WHERE filters rows; HAVING filters groups.

10. SQL Functions

Function — Definition: A reusable SQL operation that accepts values and returns a result.

Category	Examples	Purpose
String	UPPER, LOWER, TRIM, CONCAT, SUBSTRING, REPLACE, LENGTH	Text
Numeric	ROUND, CEILING, FLOOR, ABS, POWER, MOD	Numbers
Date/time	CURRENT_DATE, CURRENT_TIMESTAMP, EXTRACT	Dates/times
NULL	COALESCE, NULLIF	Missing values
Conditional	CASE	Conditional logic
Aggregate	COUNT, SUM, AVG, MIN, MAX	Summaries

COALESCE — Definition: Returns the first non-NULL expression.

NULLIF — Definition: Returns NULL when two expressions are equal; otherwise returns the first expression.

11. Date/Time SQL

Date/Time functions — Definition: Functions used to extract, compare, add, subtract and group dates/timestamps; exact syntax varies by DBMS.

Learn current date/time, year/month/day extraction, date difference, date arithmetic, timestamp truncation, time zones and month-based grouping.

12. JOINS

JOIN — Definition: Combines rows from related tables using a join condition.

JOIN	Definition	Scenario
INNER	Matching rows only	Customers with orders
LEFT	All left rows + matches	All customers including no orders
RIGHT	All right rows + matches	All orders including unmatched
FULL OUTER	All rows from both sides	Reconciliation
CROSS	Every combination	Combinations
SELF	Table joins itself	Employee-manager

JOIN	Definition	Scenario
NATURAL	Automatic same-name matching where supported	Use cautiously
LATERAL/APPLY	Right expression can reference left row where supported	Top related rows

```
SELECT c.Name, o.Order_ID, o.Amount
FROM Customers c
INNER JOIN Orders o ON c.Customer_ID=o.Customer_ID;
```

```
SELECT c.Name
FROM Customers c
LEFT JOIN Orders o ON c.Customer_ID=o.Customer_ID
WHERE o.Order_ID IS NULL;
```

13. Keys

Primary Key — Definition: Column(s) that uniquely identify each row.

Foreign Key — Definition: Column(s) that reference a key in another table.

Candidate Key — Definition: Minimal column set capable of uniquely identifying a row.

Composite Key — Definition: A key made from multiple columns.

Surrogate Key — Definition: Generated identifier such as integer or UUID.

Natural Key — Definition: A real-world attribute used as an identifier.

Alternate Key — Definition: A candidate key not selected as the primary key.

14. Constraints

Constraint — Definition: A database rule used to maintain valid and consistent data.

Constraint	Definition
NOT NULL	Value must be present
UNIQUE	Prevents duplicates subject to DBMS NULL rules
PRIMARY KEY	Uniquely identifies rows
FOREIGN KEY	Maintains relationships
CHECK	Requires a condition
DEFAULT	Supplies a default

15. Subqueries

Subquery — Definition: A query inside another SQL statement that provides a value, list or result set.

Type	Definition
Scalar	One value
Single-row	One row
Multi-row	Multiple rows
Correlated	References outer row
FROM subquery	Derived table
SELECT subquery	Value per output row

EXISTS — Definition: True when the subquery returns at least one row.

ANY/SOME — Definition: Compares a value with at least one subquery value.

ALL — Definition: Compares a value with every subquery value.

```
SELECT *
FROM Employees
WHERE Salary > (SELECT AVG(Salary) FROM Employees);
```

```

SELECT *
FROM Customers c
WHERE EXISTS(
    SELECT 1 FROM Orders o
    WHERE o.Customer_ID=c.Customer_ID
);

```

16. CTE

CTE — Definition: A named temporary result set defined with WITH for one SQL statement.

Recursive CTE — Definition: A CTE that references itself, commonly used for hierarchies and trees.

```

WITH DepartmentSalary AS(
    SELECT Department,AVG(Salary) AS Avg_Salary
    FROM Employees GROUP BY Department
)
SELECT * FROM DepartmentSalary
WHERE Avg_Salary>60000;

```

17. CASE

CASE — Definition: A conditional SQL expression similar to if/elif/else.

```

SELECT Name,
CASE WHEN Salary>=80000 THEN 'High'
      WHEN Salary>=50000 THEN 'Medium'
      ELSE 'Low' END AS Category
FROM Employees;

```

18. Set Operations

UNION — Definition: Combines compatible result sets and removes duplicates.

UNION ALL — Definition: Combines compatible results and retains duplicates.

INTERSECT — Definition: Returns common rows where supported.

EXCEPT/MINUS — Definition: Returns rows in the first result but not the second; keyword varies.

19. Part 1 Scenarios

- E-commerce: Find customers who never ordered.
- E-commerce: Calculate monthly revenue.
- E-commerce: Find the second-highest order.
- Employee: Find average salary per department.
- Employee: Find employees above company average.
- Telecom: Find devices not updated in the last 24 hours.
- Telecom: Find sites with more than 100 alarms.
- Banking: Find customers with loans but no active account.
- Data quality: Find duplicate customer records.
- AIML: Build customer features using COUNT, SUM and AVG.

20. Part 1 Interview Questions

- DQL vs DML vs DDL vs DCL vs TCL
- What is a schema?
- DELETE vs TRUNCATE vs DROP
- WHERE vs HAVING
- INNER JOIN vs LEFT JOIN
- What is a primary key?
- Primary key vs UNIQUE
- What is a composite key?
- What is a subquery?
- What is a correlated subquery?
- EXISTS vs IN
- What is a CTE?
- CTE vs subquery

- What is CASE?
- UNION vs UNION ALL
- What is NULL?